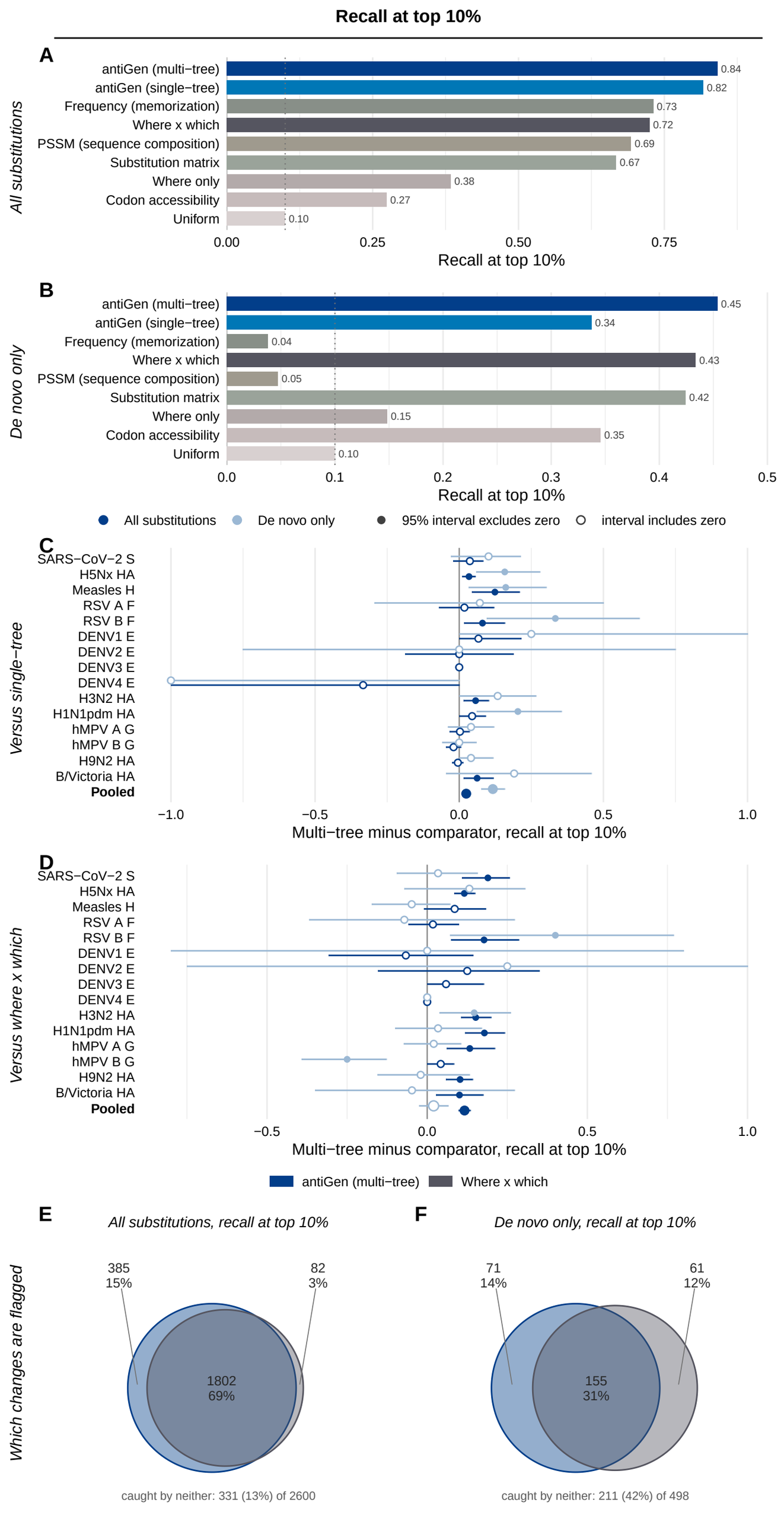
Recall at top 10%
A
All substitutions
antiGen (multi−tree)
antiGen (single−tree)
Frequency (memorization)
Where x which
PSSM (sequence composition)
Substitution matrix
Where only
Codon accessibility
Uniform
0.84
0.82
0.73
0.72
0.69
0.67
0.38
0.27
0.10
0.00
0.25
0.50
0.75
Recall at top 10%
B
De novo only
antiGen (multi−tree)
antiGen (single−tree)
Frequency (memorization)
Where x which
PSSM (sequence composition)
Substitution matrix
Where only
Codon accessibility
Uniform
0.45
0.34
0.04
0.43
0.05
0.42
0.15
0.35
0.10
0.0
0.1
0.2
0.3
0.4
0.5
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV A G
hMPV B G
H9N2 HA
B/Victoria HA
Pooled
−1.0
−0.5
0.0
0.5
1.0
Multi−tree minus comparator, recall at top 10%
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV A G
hMPV B G
H9N2 HA
B/Victoria HA
Pooled
−0.5
0.0
0.5
1.0
Multi−tree minus comparator, recall at top 10%
antiGen (multi−tree)
Where x which
E
All substitutions, recall at top 10%
F
De novo only, recall at top 10%
Which changes are flagged
385
15%
82
3%
1802
69%
caught by neither: 331 (13%) of 2600
71
14%
61
12%
155
31%
caught by neither: 211 (42%) of 498